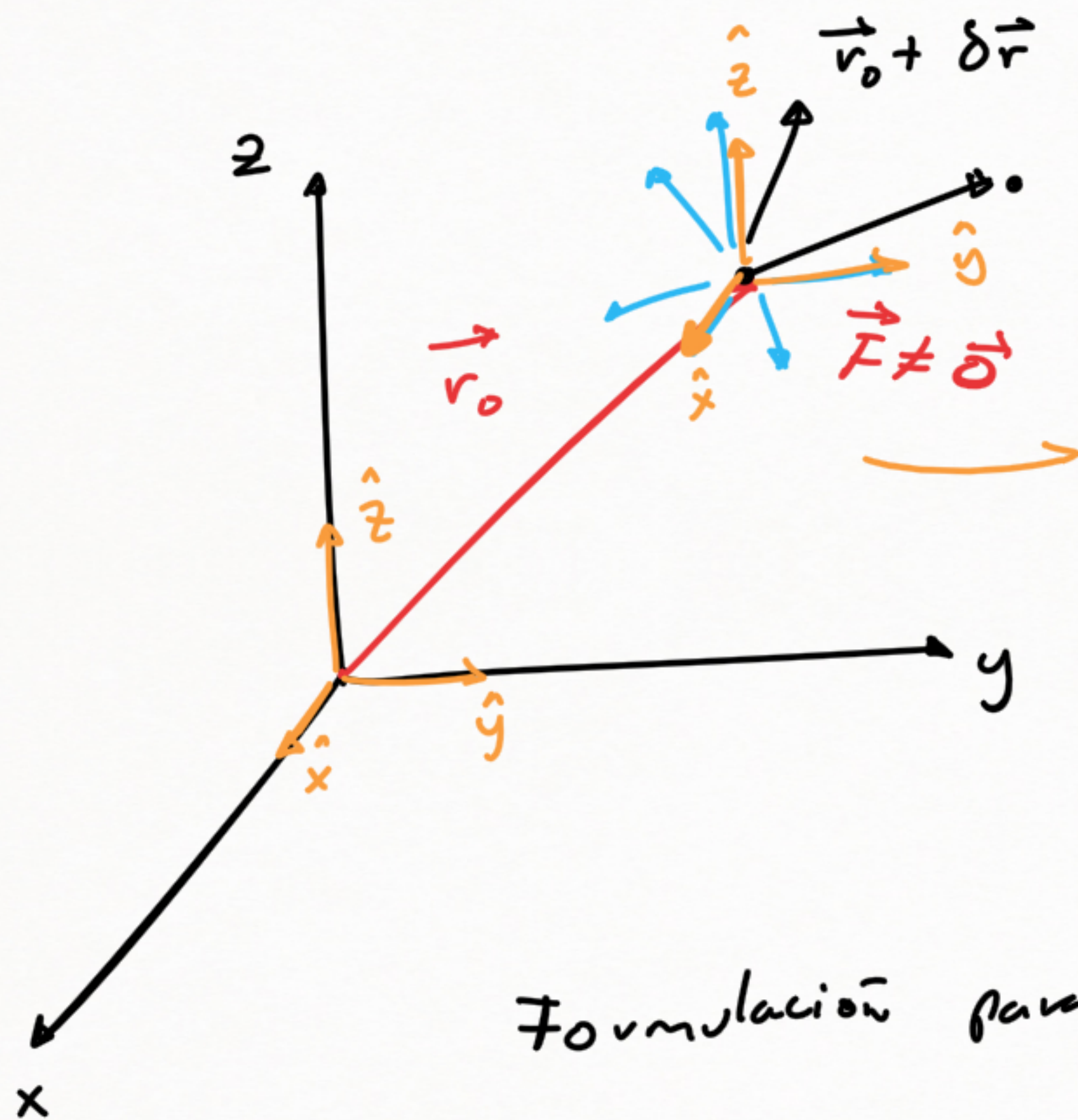




+ incluir en
forma rígida
las v. base no
deberían cambiar.

$$\delta \vec{r} = \underline{\underline{\delta x}} \hat{x} + \underline{\underline{\delta y}} \hat{y} + \underline{\underline{\delta z}} \hat{z}$$

Estos parámetros son independientes
y pueden cambiar.

Formulación para la partícula de masa m .

$$m \frac{d^2 \vec{r}}{dt^2} = \vec{F}$$

Formalismo newtoniano

$$\vec{r} = x \hat{x} + y \hat{y} + z \hat{z} ;$$

~
Ecuación vectorial a forma
escalar.

$$m \left[\frac{d^2 x}{dt^2} \hat{x} + \frac{d^2 y}{dt^2} \hat{y} + \frac{d^2 z}{dt^2} \hat{z} \right]$$

$$\{ \hat{x}, \hat{y}, \hat{z} \}$$

son v. L.I

$$= F_x \hat{x} + F_y \hat{y} + F_z \hat{z}$$

Esto es equivalente a

$$m \frac{d^2 \vec{r}}{dt^2} = \vec{F}$$

• usamos vectores "son independientes de las
coordenadas."

$$m \frac{d^2 x}{dt^2} = F_x$$

$$m \frac{d^2 y}{dt^2} = F_y$$

$$m \frac{d^2 z}{dt^2} = F_z$$

Formalismo Lagrangiano

Tenemos

$$m \frac{d^2 \vec{r}}{dt^2} - \vec{F} \cdot \delta \vec{r} = 0 \cdot \delta \vec{r} = 0$$

Tenemos

$$\left(m \frac{d^2 \vec{r}}{dt^2} \right) \cdot \delta \vec{r} = \vec{F} \cdot \delta \vec{r}$$

Trabajo

¿cuándo el trabajo resulta ser energía?

Problemas orientados cuando

$$\vec{F} = -\nabla V(\vec{r})$$

• Rotacional de $\vec{F} = 0$

$\Rightarrow$ conservativo

Integral de trayectoria es independiente del camino

Ecuaciones de Lagrange.

→ Campo conservativo

En tal caso...

$$\begin{aligned}\vec{F} \cdot \delta \vec{r} &= -\nabla U(\vec{r}) \cdot \delta \vec{r} \\ &= -\left[\frac{\partial U}{\partial x} \delta x + \frac{\partial U}{\partial y} \delta y + \frac{\partial U}{\partial z} \delta z \right]\end{aligned}$$

Desplazamientos $\{\delta j\}_{j=1}^n$ son independientes

↳ infinitesimales.

Tenemos...

$$m \frac{d^2 \vec{r}}{dt^2} \cdot \delta \vec{r} = m \left[\frac{d^2 x}{dt^2} \delta x + \frac{d^2 y}{dt^2} \delta y + \frac{d^2 z}{dt^2} \delta z \right]$$

Por tanto la s. ley de Newton queda:

$$m \frac{d^2 x}{dt^2} \delta x + m \frac{d^2 y}{dt^2} \delta y + m \frac{d^2 z}{dt^2} \delta z = - \frac{\partial U}{\partial x} \delta x - \frac{\partial U}{\partial y} \delta y - \frac{\partial U}{\partial z} \delta z$$

Energía cinética:

$$T = \frac{1}{2} m \dot{\vec{r}}^2 = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2 + \dot{z}^2) \Rightarrow \frac{\partial T}{\partial \dot{x}} = m \dot{x} \Rightarrow \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{x}} \right) = m \ddot{x}$$

Tenemos

$$\left[\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{x}} \right) + \frac{\partial U}{\partial x} \right] \delta x + \left[\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{y}} \right) + \frac{\partial U}{\partial y} \right] \delta y + \dots$$

$$+ \left[\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{z}} \right) + \frac{\partial U}{\partial z} \right] \delta z = 0$$

$$\left. \begin{aligned} \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{x}} \right) + \frac{\partial U}{\partial x} &= 0 \\ \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{y}} \right) + \frac{\partial U}{\partial y} &= 0 \\ \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{z}} \right) + \frac{\partial U}{\partial z} &= 0 \end{aligned} \right\} \begin{array}{l} 2^{\text{da}} \text{ lei de Newton para escalares} \end{array}$$

Tenemos el siguiente escalar:

$$\mathcal{I}(\vec{r}, \dot{\vec{r}}, t) = T - U(\vec{r})$$

$$\frac{\partial T}{\partial \dot{x}} = \frac{\partial \mathcal{I}}{\partial \dot{x}}; \quad \frac{\partial T}{\partial \dot{y}} = \frac{\partial \mathcal{I}}{\partial \dot{y}}; \quad \frac{\partial T}{\partial \dot{z}} = \frac{\partial \mathcal{I}}{\partial \dot{z}}$$

$$\frac{\partial U}{\partial x} = - \frac{\partial \mathcal{L}}{\partial x}; \quad \frac{\partial U}{\partial y} = - \frac{\partial \mathcal{L}}{\partial y}; \quad \frac{\partial U}{\partial z} = - \frac{\partial \mathcal{L}}{\partial z}$$

$$T = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2 + \dot{z}^2)$$

$$x = x_1$$

$$y = x_2$$

$$z = x_3$$

